



- 2 The cubic equation $x^3 + bx^2 + cx + d = 0$, where $d \neq 0$, has roots α, β, γ such that $\gamma = \frac{1}{\beta}$.

(a) Show that $\beta + \frac{1}{\beta} = d - b$.

[3]

This image shows a full page of white paper with horizontal dashed lines, typical of primary school handwriting practice paper. The lines are evenly spaced and run across the entire width of the page. There are no margins, text, or other markings present.

(b) Show also that $\beta + \frac{1}{\beta} = \frac{1-c}{d}$.

[2]

[illegible]



(c) It is given that $b = 3$, $c = -3$ and $d > 0$.

(i) Find the value of d .

[2]

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(ii) Find the value of $\alpha^2 + \beta^2 + \gamma^2$.

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